

Supervised Bayesian Matrix Factorization Identifies Prognostic Gene Expression Programs

Project Update

Andrew Walther

Department of Biostatistics, UNC Gillings School of Global Public Health

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Agenda

Background · The PDAC prognostic problem · why supervise the factorization

Method updates · The joint model · two linear predictors ($L\beta$ vs $(YF)\beta$) · explicit priors & non-parametric baseline hazard · the ELBO & variational family · CAVI updates & convergence · preprocessing · selecting K · the cohort-membership extension

Results · Multi-cohort simulation study · real-data gene programs & survival · external validation on five held-out cohorts

Impact & Future Directions

Background

Background: PDAC needs molecular prognosis

- Five-year survival < 12% — among the most lethal solid tumors
- Most patients diagnosed late; **no widely adopted molecular stratification** at diagnosis
- Bulk expression mixes malignant, stromal, and immune signals — prognostic structure is interwoven and often **low-variance**
- **Goal:** recover *gene expression programs (GEPs)* whose activation predicts survival, and validate them on **independent** cohorts
- A GEP is a coordinated set of genes (a column of F) whose patient-level activation (a column of L) we test for survival association

Why model expression and survival *jointly*?

The common pipeline is **two-step**: run unsupervised factor analysis (PCA / EBMF), then test factors for survival *post hoc*.

- Unsupervised axes track the **largest expression variance** — often batch or housekeeping signal
- A program explaining little variance but strongly prognostic gets **missed**
- **Supervision** steers factors toward survival-relevant structure *during* estimation

A program explaining 15% of variance may be pure batch effect; one explaining 3% may cleanly separate survivors. Supervision prioritizes the second.

Methods

The model — two coupled components

$$\underbrace{Y = LF^\top + E, \quad E_{ij} \sim N(0, \tau_j^{-1})}_{\text{genomics: low-rank+gene-specific noise}} \quad \underbrace{h_i(t) = h_0(t) \exp(\eta_i)}_{\text{Cox proportional hazards}}$$

Quantity	Dim	Meaning
Y	$n \times p$	expression matrix
L	$n \times K$	patient program-activation scores
F	$p \times K$	gene weights per program
β	$K \times 1$	survival (log-HR) coefficients
τ_j	scalar	gene-specific noise precision
η_i	scalar	patient log-risk score ($L_i\beta$ or $(Y_i F)\beta$)

Shared loadings couple the two

The **same** latent structure drives low-rank reconstruction of Y **and** the survival risk score. A program is **prognostic only** if its $\beta_k \neq 0$; otherwise it explains expression alone.

Non-parametric baseline hazard

$h_0(t)$ is left **unspecified** — it cancels in the Cox partial likelihood. No Weibull / exponential shape assumption; only proportional hazards.

Two linear predictors: $L\beta$ vs $(YF)\beta$

Loadings predictor — $\eta = L\beta$

- Risk uses the **learned** patient scores L directly
- Survival shapes L during estimation (fully joint)
- Scoring a **new** patient requires **re-deriving** their L (an OLS projection) — an **approximation** of the trained scores

Projection predictor — $\eta = (YF)\beta$

- Risk uses the **direct genomic projection** YF
- **Exact** out-of-sample scoring: $\eta_{\text{new}} = (Y_{\text{new}}F)\beta$ — the *identical* formula at train and test, no train/test score mismatch

Priors — every block is explicit. $q(\cdot)$ denotes the **variational (approximate posterior)** for each block, fit by empirical Bayes at every update:

- $q(L_{\cdot k}), q(F_{\cdot k})$ — **point-exponential** (non-negative spike-and-slab):
 $g(\theta) = \pi_0 \delta_0 + (1 - \pi_0) \text{Exp}(\lambda), \theta \geq 0$
- $q(\beta_k)$ — **normal** (Gaussian slab, no spike): $g(\beta) = N(0, \sigma^2)$
- $q(\tau_j)$ — **closed-form MLE**: no EBNM, no prior tuning

Sparsity π_0 , scale λ , and slab variance σ^2 are all estimated from the data (`ebnm`) at each step.

Why we report both

$(YF)\beta$ gives leakage-free **exact** external scoring → the recommended model; $L\beta$ is the fully-joint reference.

The objective: ELBO & variational family

Mean-field variational family — each block factorizes independently:

$$q(L, F, \beta, \tau) = \prod_{k=1}^K q(L_{\cdot k}) \prod_{k=1}^K q(F_{\cdot k}) \prod_{k=1}^K q(\beta_k) \prod_{j=1}^p q(\tau_j)$$

The **ELBO** (evidence lower bound) is the objective CAVI maximizes — a lower bound on the log marginal likelihood. For the recommended **projection** model:

$$\text{ELBO} = \underbrace{E_q[\mathcal{L}_{\text{gen}}]}_{\text{expression fit}} + \underbrace{\lambda E_q[\mathcal{L}_{\text{Cox}}]}_{\text{survival fit}} - \underbrace{\sum_{\text{blocks}} D_{\text{KL}}(q \parallel \text{prior})}_{\text{prior regularization}}$$

- **Genomics fit** $E_q[\mathcal{L}_{\text{gen}}] = \sum_j E_q \log p(Y_{\cdot j} \mid L, F_j, \tau_j)$ — Gaussian reconstruction of Y by $LF^\top$
- **Survival fit** $E_q[\mathcal{L}_{\text{Cox}}]$ — a **2nd-order Taylor** expansion of the Cox partial log-likelihood in $\eta = (YF)\beta$, making the survival term conjugate to the EBNM updates
- **KL term** penalizes each posterior q away from its prior — the adaptive shrinkage
- The **shared L** enters both likelihood terms — that coupling *is* the supervision
- The ELBO **balances** expression and survival automatically — **no tuning parameter α** ($\lambda = 1$ in the recommended fits)

*What the ELBO is **for**: it is the **fitting objective** — CAVI maximizes it, and a fit is declared converged when it stops improving. It is **not** used to choose the number of programs — that is done by cross-validated C-index (next: **Selecting K**). ::: {notes} The variational family is mean-field and factor-wise, written out at the top — a direct April request. The ELBO has three pieces: Gaussian genomics reconstruction, the Cox survival term via a second-order Taylor expansion that keeps the updates conjugate, and a per-block KL penalty that is the adaptive shrinkage. I show the survival term for the projection $(YF)\beta$ model, which is the recommended one. The shared L in both likelihood terms is the supervision, and the ELBO balances the two with no α to tune. :::*

Inference: factor-wise CAVI

Coordinate ascent (Gauss–Seidel) — per outer iteration, for each factor k :

$$\beta_k \rightarrow L_{\cdot k} \rightarrow F_{\cdot k}, \quad \text{then } \tau \text{ once per sweep}$$

- Each update is an **EBNM** sub-problem: pseudo-data $x = B/A$, scale $s = 1/\sqrt{A}$, then `ebnm` returns the posterior mean **and** variance
- L has a **dual-source precision** (genomics + survival):

$$A_{L,i} = \underbrace{\sum_j \tau_j E[F_{jk}^2]}_{\text{expression}} + \underbrace{W_{ii} E[\beta_k^2]}_{\text{survival}}$$

where W_{ii} is the patient's Cox working weight

- The empirical-Bayes prior $\hat{g}$ is **re-estimated every update** → adaptive shrinkage per factor / per gene; deflate the residual after each factor

Convergence — in plain English. After a 5-iteration burn-in, stop when the **ELBO stops improving** — its relative change between iterations falls below 10^{-5} :

$$\frac{|\text{ELBO}^{(t)} - \text{ELBO}^{(t-1)}|}{|\text{ELBO}^{(t-1)}|} < 10^{-5}$$

- The ELBO is the **single objective** being maximized, so this directly detects a stationary point of the variational optimization
- Mean absolute changes in L and β are tracked alongside as **diagnostics** (both settle below 10^{-3} at convergence)



Risk-direction sign correction

Cox concordance is invariant to a global sign flip, so an unconstrained fit can land at $C < 0.5$. **Fix:** orient each program against the **training** concordance, then apply the *same* orientation at test. (Used to define program activation in the survival results.)

Preprocessing is crucial: per-platform normalization

The $p \gg n$ scaling problem. Merging platforms (RNA-seq, microarray, proteomics) leaves each gene on a different scale. The survival precision for program k is

$$A_k = \sum_i w_i (Y F)_{ik}^2,$$

where w_i = the patient's **Cox working weight** (curvature from the survival Taylor step) and $(Y F)_{ik}$ = patient i 's **projection score** on program k . Without correction, between-platform variance inflates $(Y F)$ unevenly, **swamps** A_k , and the survival coefficients **collapse to** $\beta \rightarrow 0$.

The fix — normalize *per platform, before merging*:

1. Log2 transform (RNA-seq only)
2. **Per-cohort gene selection** (combined rank of mean $\times$ variance), top-3000 per cohort, then **intersect** $\rightarrow$ **2064 genes** in the merged training matrix
3. **Per-platform z-standardization** (each cohort centered/scaled on its own)
4. Row-bind into the merged training matrix

⚠ Skip per-platform normalization $\rightarrow \beta$ collapses

In an 18-configuration benchmark, **10 of 12** pipelines that do **not** z-standardize per platform before merging drive $\beta \rightarrow 0$ — for *both* linear predictors.

💡 No leakage

Normalization is fit **within each training cohort, before** any split; held-out cohorts get the *same* per-cohort recipe at scoring time.

Selecting the number of programs: K and K_{eff}

Two distinct filters — the deck keeps them separate:

1 · Retained programs K — by cross-validation

- **5-fold CV** on the training data; record the held-out **C-index** at each K in a grid
- **1-SE rule**: pick the smallest K whose mean CV C-index is within one standard error of the best
- **Biological floor** $K \geq 3$: a *programs* comparison needs at least three (aligned with the lab's penalized survival-MF work)
- $\Rightarrow$ recommended preprocessing selects **$K = 7$**

2 · Survival-active programs K_{eff} — by $|\hat{\beta}|$

- Fit the model at the retained $K = 7$
- A program is **survival-active** if its posterior coefficient clears the activity threshold: $|\hat{\beta}_k| > 10^{-3}$
- The rest have $\hat{\beta}_k$ shrunk to ≈ 0 — they **explain expression only**
- $\Rightarrow$ **$K_{\text{eff}} = 2$** of the 7 retained programs carry survival signal

 Read both numbers together

K is *how many programs the model keeps* (reconstruction; chosen by CV). K_{eff} is *how many of those are prognostic* (chosen by $|\hat{\beta}|$). For the recommended model: $K = 7 \rightarrow K_{\text{eff}} = 2$.

L extension: cohort-membership indicator

Idea. Append corner-point dummy columns (one per non-reference cohort) to L , with their survival coefficient **fixed at** $\beta_{\text{cohort}} \equiv 0$, so survival risk is driven only by the **biological** programs while the dummies absorb platform offsets in the genomics term.

- Updated via a Normal-conjugate step (cohort-row F update)
- The forced dummy consumes **one degree of freedom**, so we select **one additional factor** ($K = 7 \rightarrow 8$) to keep the same number of biological programs

When it helps — and when it doesn't

- At low K with weak normalization, the indicator can rescue platform-confounded fits
- Under **per-platform z-standardization**, the offset is *already* removed — the indicator adds a factor slot for **no benefit**

Recommendation

With per-platform z-standardization, adding the cohort indicator **moves mean external C-index** *down* from 0.636 to 0.614 (and $K : 7 \rightarrow 8$).

We therefore recommend *no* cohort indicator — normalization already absorbs the platform offset.

Results

Data — Synthetic Validation & PDAC Cohorts

Multi-cohort simulation

EBMF-grounded ground truth: shared factors (present in all studies) + study-specific factors; **shared factors carry the survival signal**, study-specific do not.

Scenarios

- *All Shared Factors* — every program is shared & prognostic
- *All Cohort-Specific Factors* — negative control; no shared signal ($\eta \equiv 0$)
- *Shared & Cohort-Specific Factors* — the realistic mix

Model Arms

- EBMF (unsupervised two-step baseline)
- $\text{YFB_base} \cdot \text{YFB_cohort} \cdot \text{LB_base} \cdot \text{LB_cohort}$

PDAC cohorts — External Validation

- **Train:** TCGA-PAAD + CPTAC ($n \approx 273$, RNA-seq + proteomics)
- **Validate:** 5 fully held-out cohorts, scored by the **exact** projection formula

$$\eta = (Y_{\text{new}} F) \beta$$

Held-out cohort	Platform
Dijk	RNA-seq
Moffitt	Microarray
PACA-AU (array)	Microarray
PACA-AU (seq)	RNA-seq
Puleo	Microarray

Synthetic Results — Shared & Cohort-Specific Factor Recovery

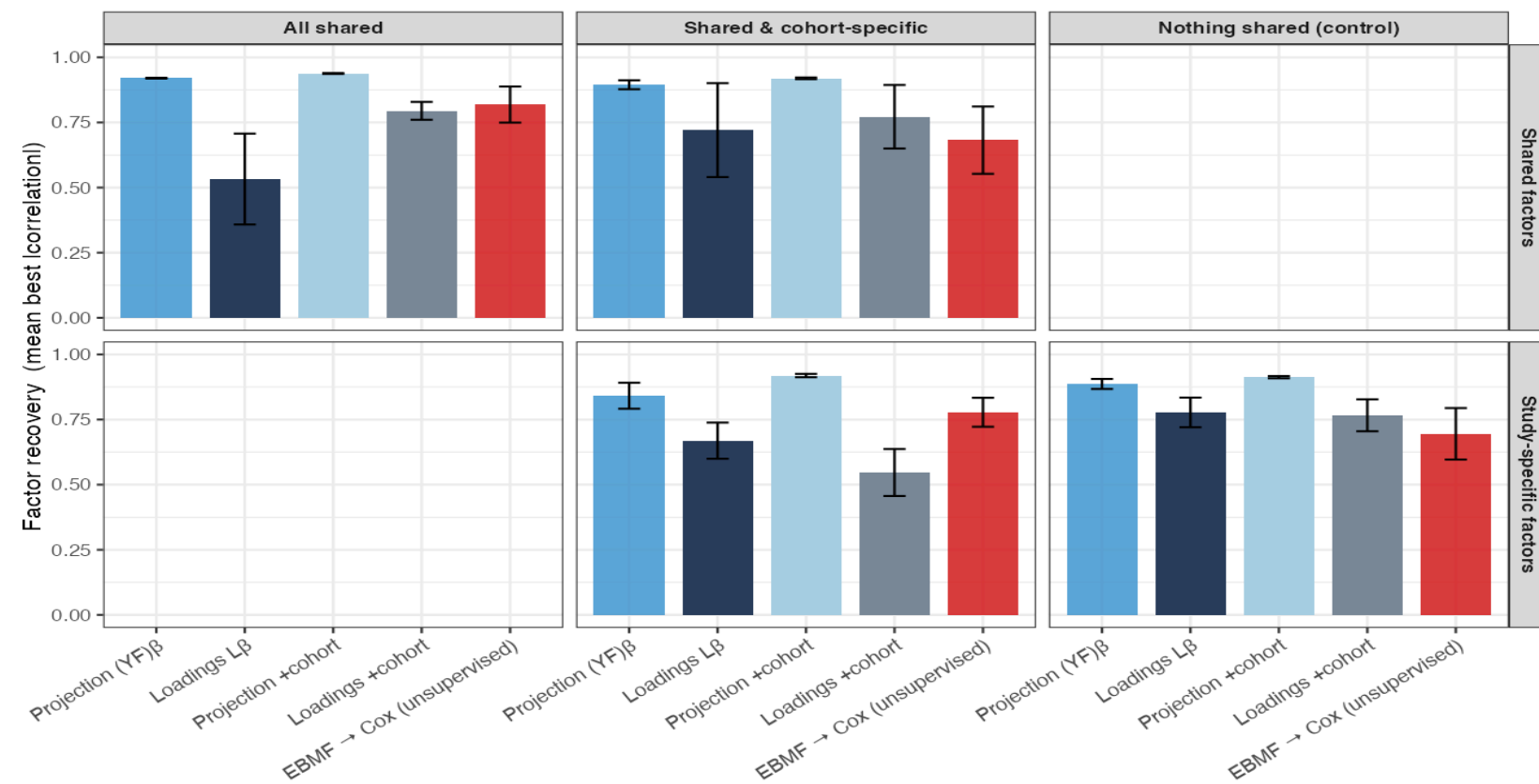


Figure 1. Factor recovery = mean best $|correlation|$ between each estimated gene program and the true program it matches, by model arm and scenario (mean $\pm$ SE over 5 seeds). Top row: shared (prognostic) factors; bottom row: study-specific factors.

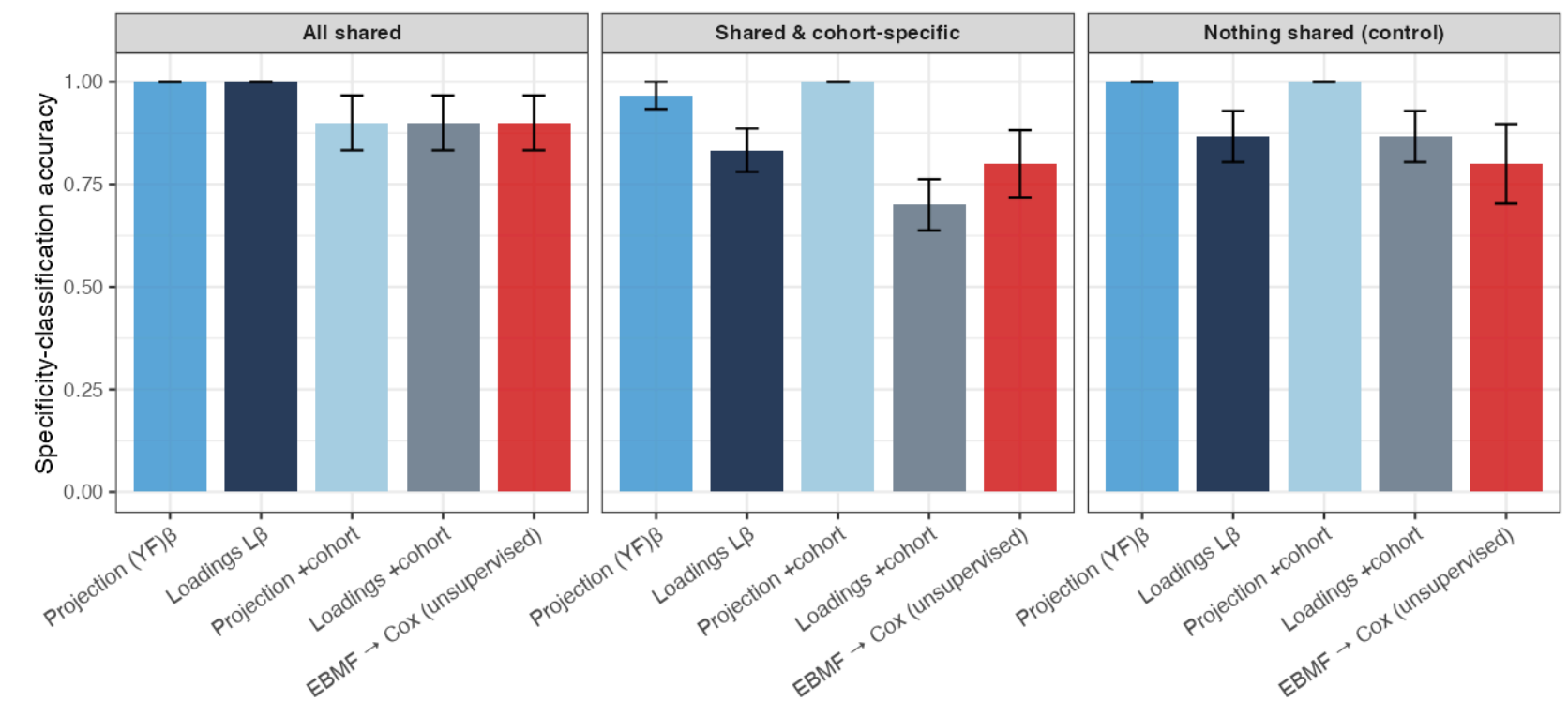


Figure 2. Specificity-classification accuracy = fraction of recovered factors correctly *typed* as shared vs. study-specific, from per-cohort loading energy alone (mean $\pm$ SE, 5 seeds).

- **Two complementary questions.** Figure 1 asks *how faithfully* each gene program is reconstructed (a correlation); Figure 2 asks whether the model *correctly labels* each program as shared or study-specific (a classification). High recovery with low typing accuracy would mean good programs, wrong structure — so we report both.
- The **projection model** recovers shared factors at $|r| \approx 0.90\text{--}0.92$ and types them with accuracy up to **1.00** — it natively separates shared from study-specific structure, which is *why* the cohort indicator adds little.

Synthetic Results — Factor Recovery Heatmap

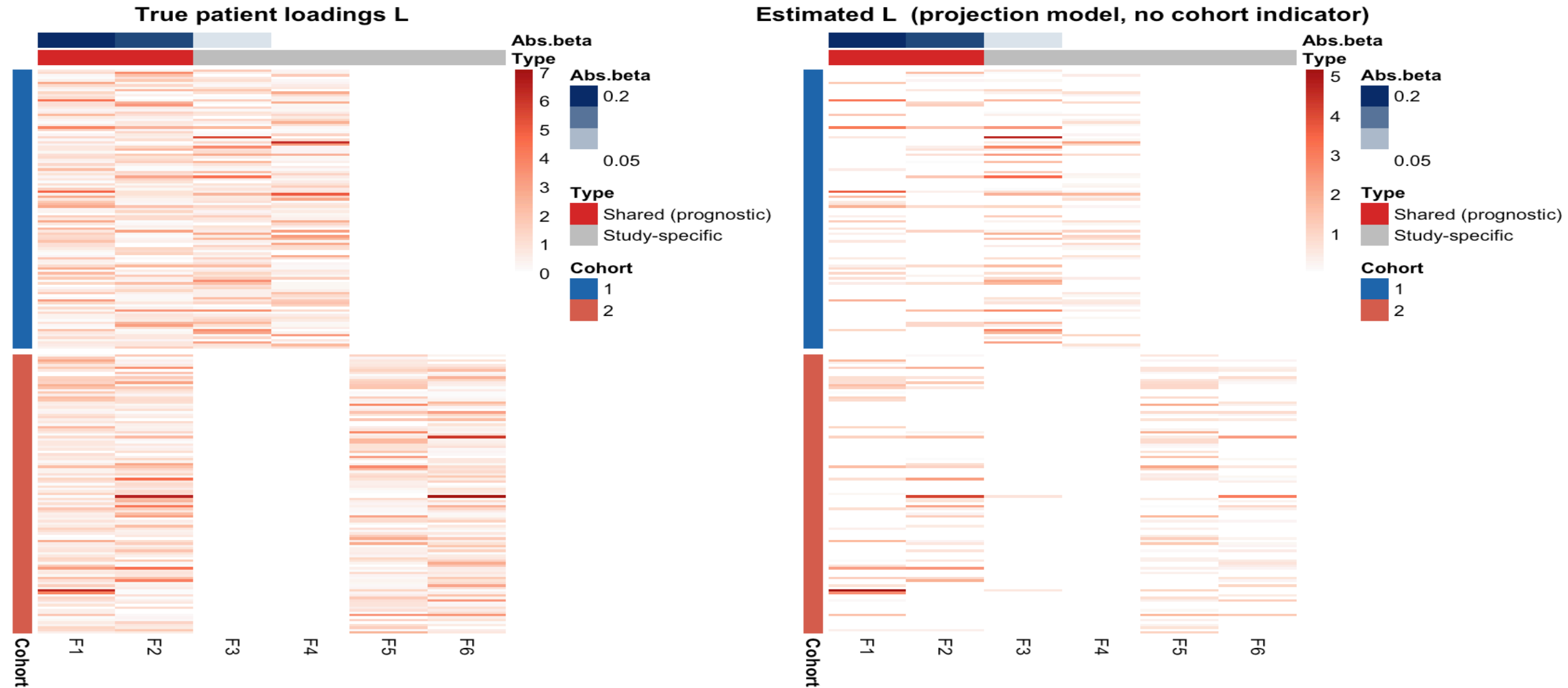
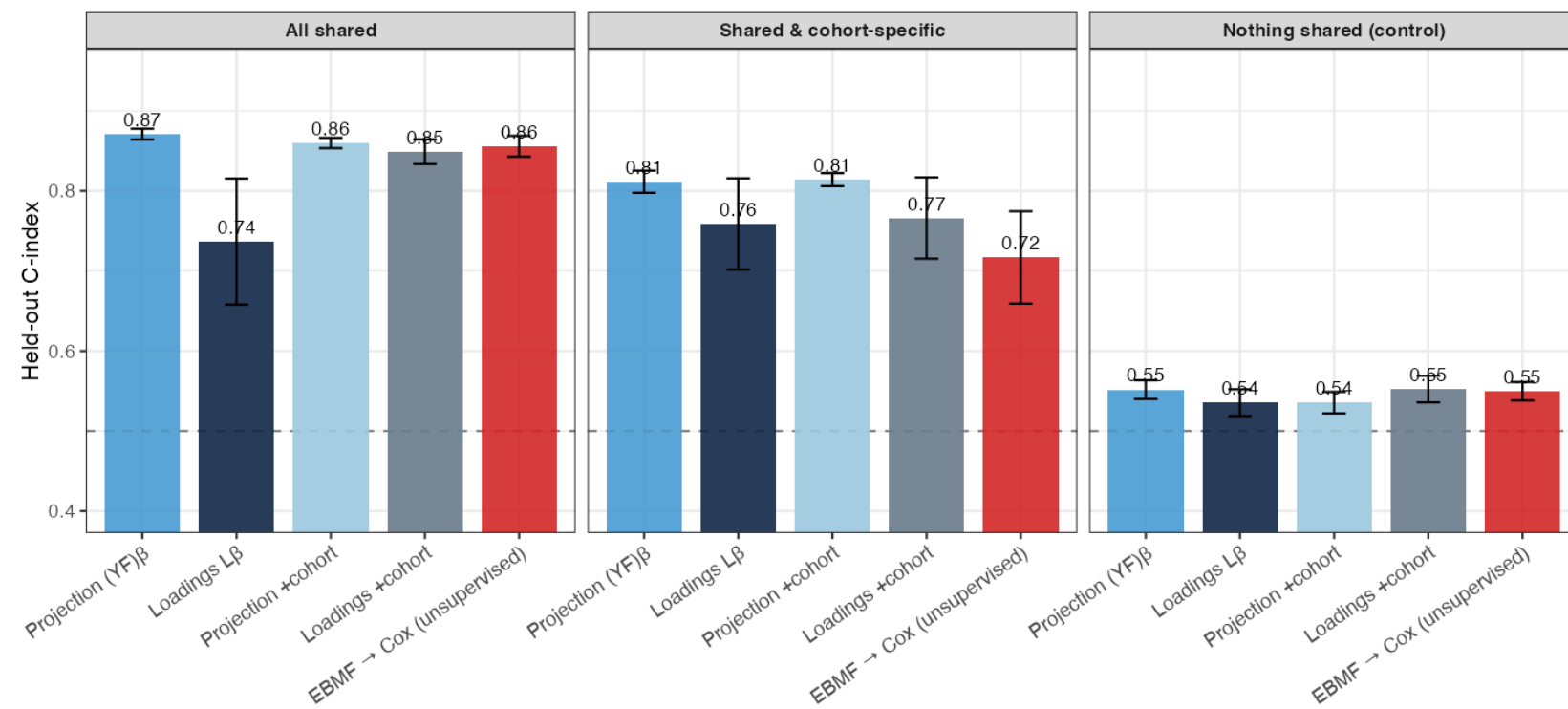


Figure 3. Patient-loading matrix L in the realistic *Shared & Cohort-Specific* scenario (hybrid, seed 1). Rows = patients grouped by cohort (left bar); columns = factors. Top annotations: true factor type (shared/prognostic vs. study-specific) and $|\hat{\beta}|$. **Left:** true block-sparse L . **Right:** estimated L from the projection model — recovering shared factors (active in *both* cohorts) and study-specific factors (active in one), with **no cohort indicator**.

- Shared factors light up across **both** cohort blocks; study-specific factors light up in **one** — the model recovers the intended structure directly from the data.

Synthetic Results — SSBMF outperforms EBMF+Cox



Hybrid (Shared & Cohort-Specific), mean C-index (over 5 seeds):

- Projection (YF) β : 0.81
- Loadings $L\beta$: 0.76
- Unsupervised EBMF → Cox: 0.72

Negative control (All Cohort-Specific): all arms at ≈ 0.54 — no false prognostic signal manufactured when none exists.

Figure 4. Held-out C-index by model arm within each scenario (mean $\pm$ SE over 5 seeds; dashed line = chance). The unsupervised EBMF $\rightarrow$ Cox baseline is shown alongside the supervised arms.

Takeaway

Joint supervision beats the unsupervised two-step approach on data with known ground truth — and invents no signal when there is none.

PDAC Cohorts — Prognostic Gene Programs

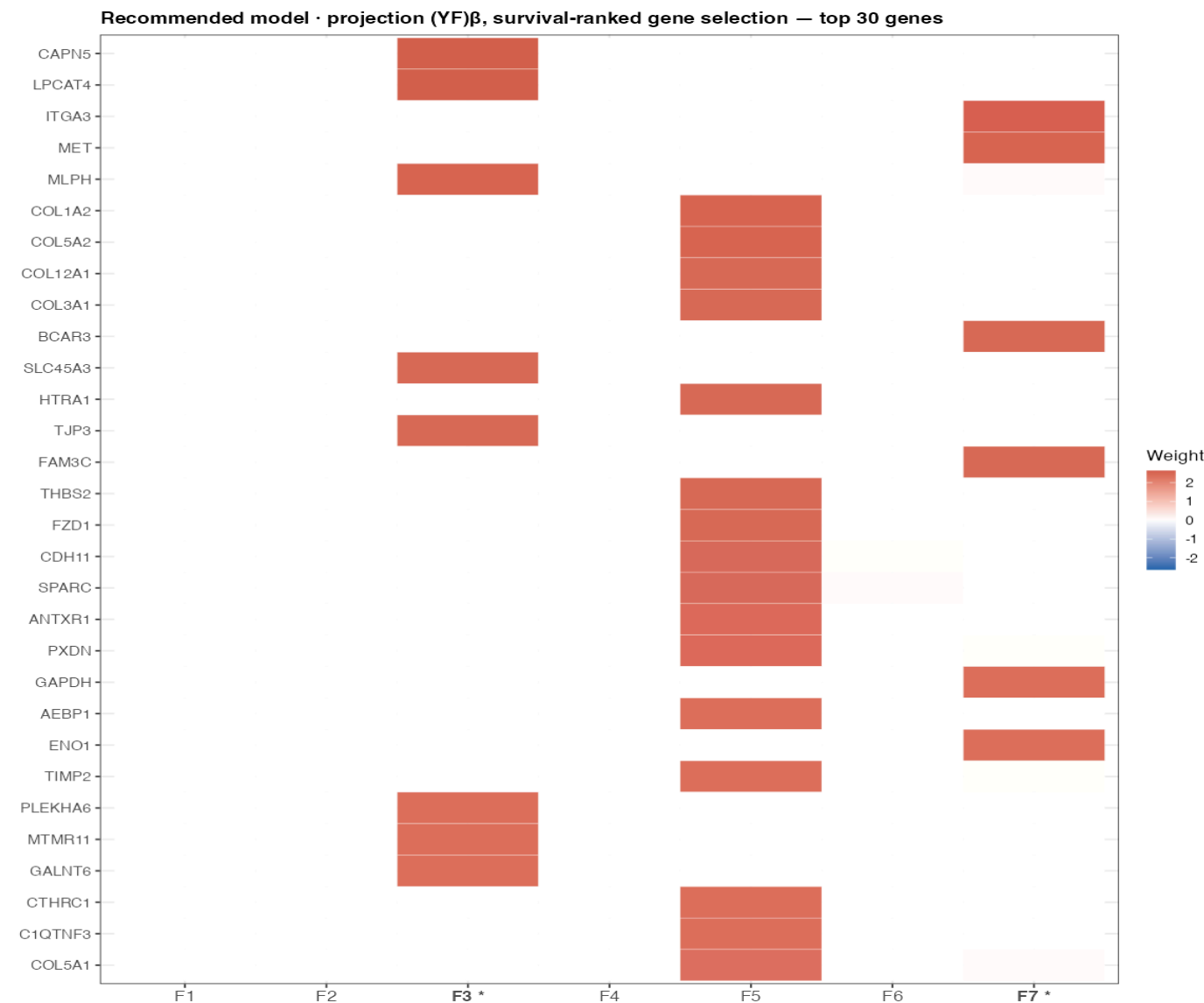


Figure 5. Gene-weight heatmap (top 30 genes) for the **recommended model**: projection $(YF)\beta$, per-platform z-standardization, survival-ranked gene selection (2064 genes), **no** cohort indicator, $K = 7$. Weights are **non-negative** (point-exponential prior). Survival-active programs ($|\hat{\beta}| > 10^{-3}$) are starred.

- $K = 7$ programs **retained** (CV + 1-SE, floor $K \geq 3$); $K_{\text{eff}} = 2$ **survival-active** — Program 3 (**protective**) and Program 7 (**adverse**)
- The *largest* expression program (collagen/ECM/stromal) is **not** survival-active — supervision separates expression-only structure from prognostic structure

PDAC Cohorts — Prognostic Association

The two survival-active programs stratify patients (split at median **projection score** $(YF)_k$ — the quantity the model scores on):

- **Adverse — Program 7** (higher activation → shorter survival), log-rank $p = 0.00047$
 - top genes: *ITGA3*, *MET*, *BCAR3*, *FAM3C*, *GAPDH*, *ENO1*, *TPI1*, *PGK1*
- **Protective — Program 3** (higher activation → longer survival), log-rank $p = 0.0077$
 - top genes: *CAPN5*, *LPCAT4*, *MLPH*, *SLC45A3*, *TJP3*, *PLEKHA6*, *MTMR11*, *GALNT6*

Proportional-hazards assumption not violated (Schoenfeld test).

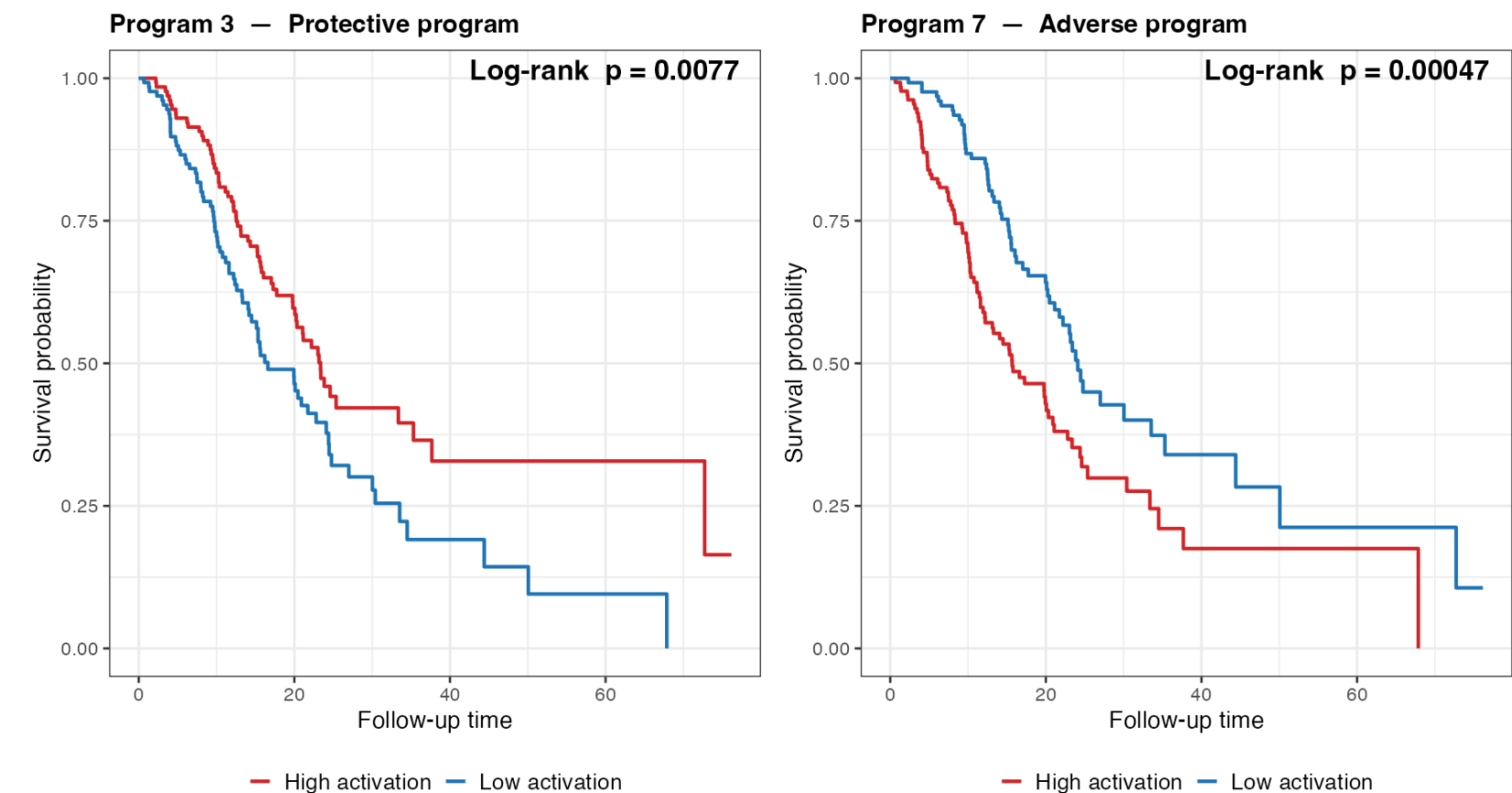


Figure 6. Kaplan–Meier curves for the two survival-active programs, patients split at the median projection score $(YF)_k$ in the training cohorts. The adverse program’s high-activation group has worse survival; the protective program’s, better. Direction read empirically from the curves.

PDAC Cohorts — External C-index

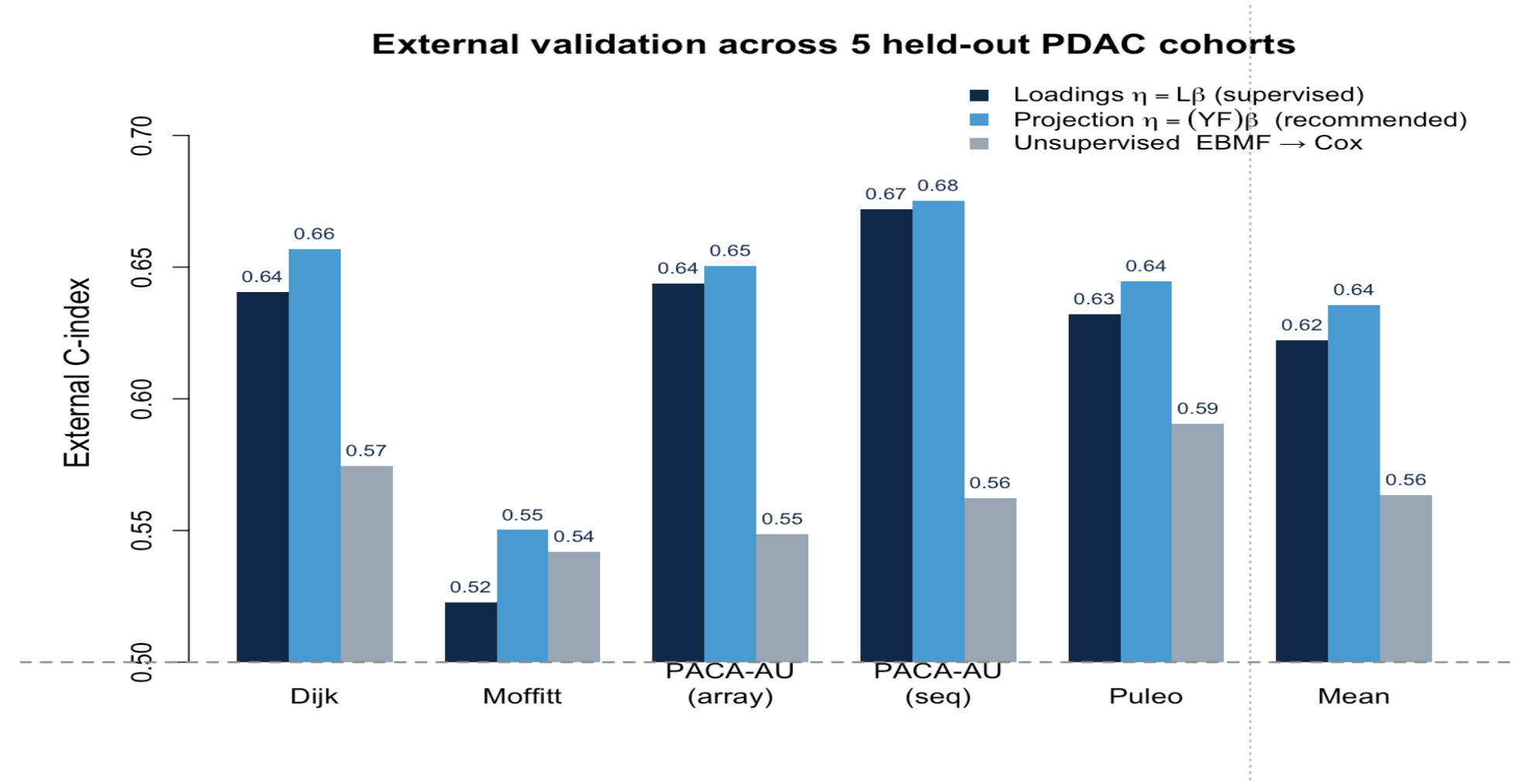


Figure 7. External C-index across the five held-out PDAC cohorts (+ mean) for three arms: loadings $L\beta$, projection $(YF)\beta$, and the unsupervised EBMF $\rightarrow$ Cox two-step. Dashed line = chance.

- **Projection (YFB) model mean C = 0.636** · loadings (LB) model 0.622 · EBMF $\rightarrow$ Cox 0.564, across the five held-out cohorts
- The projection-based supervised joint model **outperforms the unsupervised two-step approach across all 5 external PDAC cohorts**, with $K = 7$ (CV + 1-SE, floor $K \geq 3$) $\rightarrow K_{\text{eff}} = 2$

Model Comparison Summary

Model	Linear predictor	Preprocessing	Cohort ind.	K	K_eff	Synthetic C	Real ext. C
Projection (recommended)	$(YF)\beta$	per-platform z-std + survival-ranked gene selection	no	7	2	0.81	0.636
Loadings	$L\beta$	per-platform z-std + survival-ranked gene selection	no	7	2	0.76	0.622
Projection + cohort ind.	$(YF)\beta$	per-platform z-std + survival-ranked gene selection	yes	8	2	0.81	0.614
Unsupervised two-step	EBMF $\rightarrow$ Cox	per-platform z-std + survival-ranked gene selection	n/a	20	n/a	0.72	0.564

Synthetic C = mean held-out C-index, hybrid scenario (5 seeds). Real ext. C = mean across 5 held-out PDAC cohorts. **K** = CV + 1-SE selected (biological floor $K \geq 3$); **K_eff** = survival-active programs ($|\hat{\beta}| > 1e-3$). Gene selection ranks genes by combined mean $\times$ variance per cohort.

- **Recommendation:** projection predictor $(YF)\beta$ · per-platform z-standardization · survival-ranked gene selection · **no** cohort indicator
- Wins on both axes — best synthetic **and** real external C-index — with the most **parsimonious, interpretable** survival structure ($K = 7 \rightarrow K_{\text{eff}} = 2$) and **exact** external scoring



On gene selection

The combined-rank selection recipe was **motivated by** the lab’s penalized survival-MF (DeSurv) preprocessing; here it is applied within the fully Bayesian model.

Impact / key findings

1. **Joint supervision beats the unsupervised two-step** — by ~0.09 C-index on synthetic data with known truth (0.81 vs 0.72, hybrid), and across **all five** held-out PDAC cohorts on real data.
2. **Preprocessing is decisive** — per-platform z-standardization before merging is the difference between a working model and $\beta \rightarrow 0$; **10 of 12** alternatives collapse.
3. **Parsimony with signal** — the recommended model carries just $K_{\text{eff}} = 2$ **survival-active programs** (vs ~20 unsupervised factors) yet matches the field, with exact, leakage-free external scoring.
4. **Mean external C-index = 0.636** across five independent PDAC cohorts — fully Bayesian throughout (adaptive priors, no α tuning, uncertainty-aware).

Limitations

- **Small external cohorts give noisy estimates** — per-cohort C ranges down to 0.55 (Moffitt), near chance
- **Real-data margin over unsupervised EBMF $\rightarrow$ Cox is modest** (EBMF is already ≈ 0.63 within-training); the *synthetic* gap (0.81 vs 0.72) is the cleaner demonstration of supervision's value
- The two programs are **statistically prognostic** — **biological annotation (pathway enrichment) is the next step**, not yet done

Future directions & summary

Addressed since April

- CV-based K selection ✓ — 5-fold CV, 1-SE rule
- Multi-cohort shared-factor validation ✓ — completed simulation study

Next steps

1. **Pathway enrichment** on the two survival-active programs — assign biology to the statistical factors
2. **Manuscript preparation** — methods-primary framing; PDAC as the application

Summary

1. Joint supervised factorization, two linear predictors; **point-exponential priors on L , F , a normal prior on β** , non-parametric baseline hazard
2. **Preprocessing is crucial** — per-platform normalization prevents the survival precision from being dominated and β from collapsing
3. Validated on a completed multi-cohort simulation (hybrid C = 0.81 vs 0.72) **and** five external PDAC cohorts (mean C = 0.636)
4. **Recommended model**
 - Linear predictor: projection $(Y F)\beta$
 - L cohort indicator: No
 - Pre-processing: per-platform z-std
 - Selected $K = 7$ ($K_{\text{eff}} = 2$), mean external C = **0.636**

Discussion

Thank You

Questions, suggestions, and feedback welcome!

Andrew Walther · awalther@unc.edu

Department of Biostatistics, UNC Gillings School of Global Public Health

Appendix

Appendix — cohort-indicator contrast

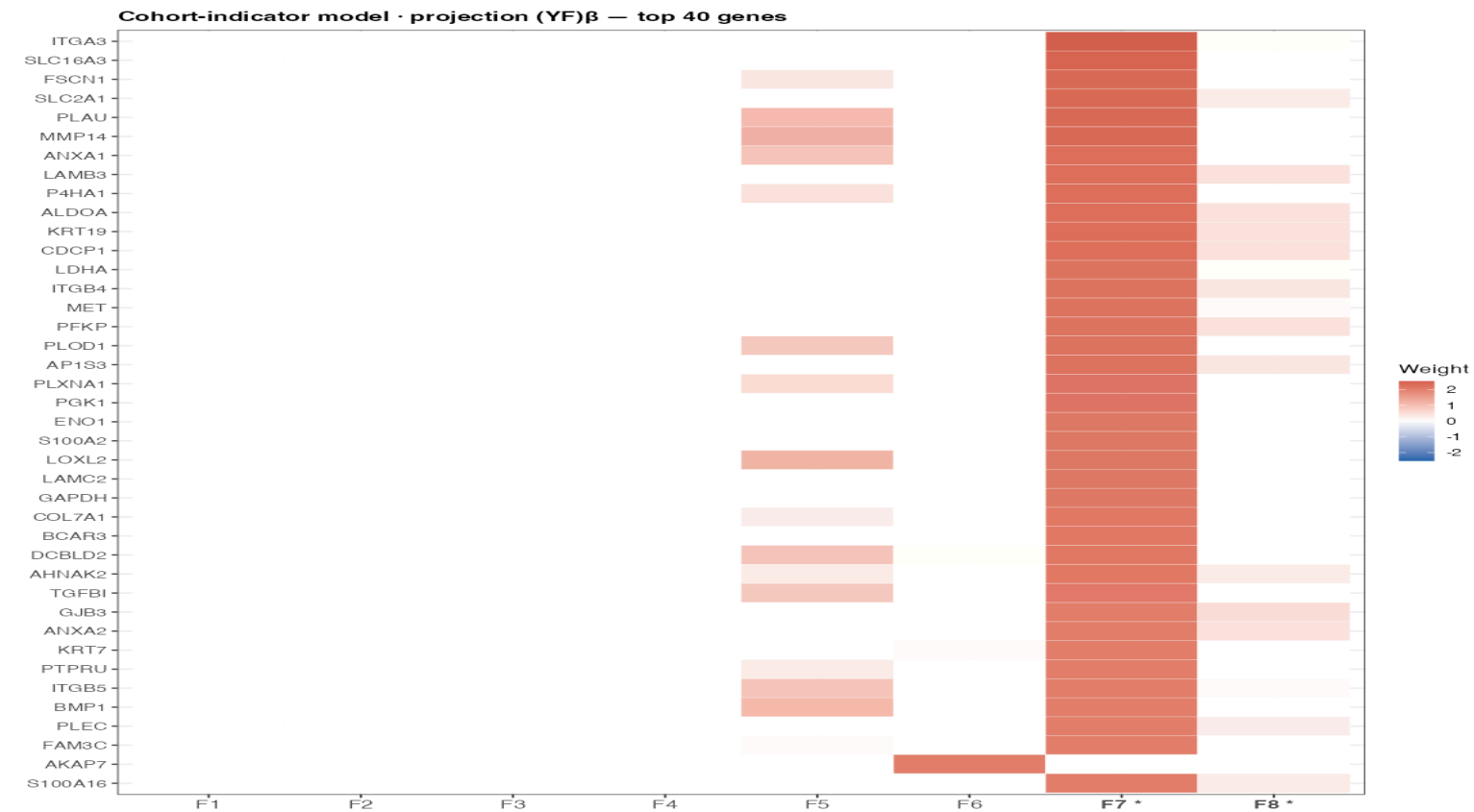


Figure A1. Gene-weight heatmap with the cohort-membership indicator added (projection model, $K = 8$). Survival-active programs starred.

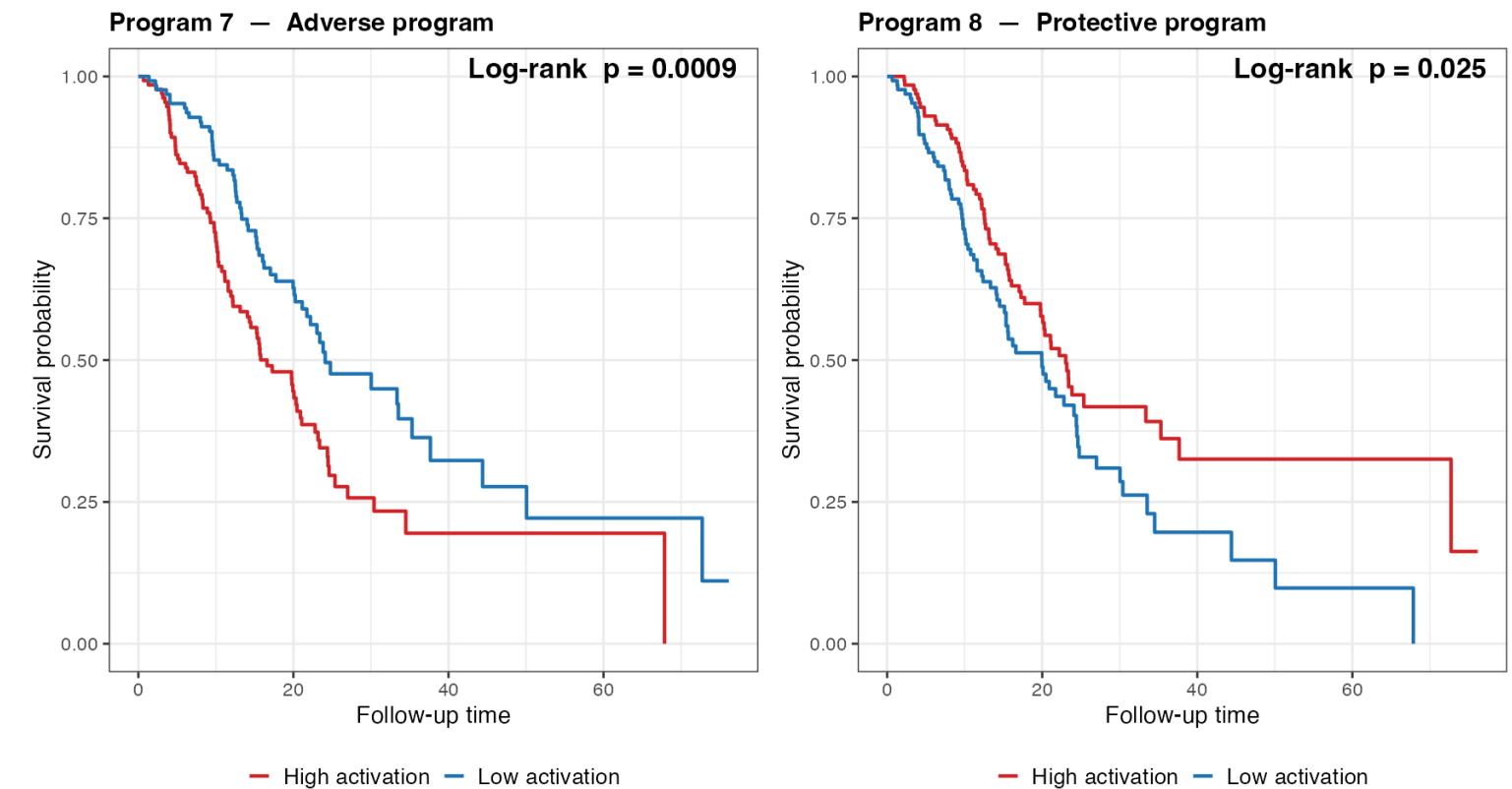


Figure A2. Survival stratification with the cohort-membership indicator, by projection score. The survival-active structure is essentially unchanged from the recommended model.

- Adding the indicator under per-platform z-std spends a factor slot ($K : 7 \rightarrow 8$) **without** improving external C-index ($0.636 \rightarrow 0.614$)
- The biological survival-active structure is essentially unchanged — confirming the platform offset is already removed by normalization

Appendix — convergence & survival-activity magnitudes

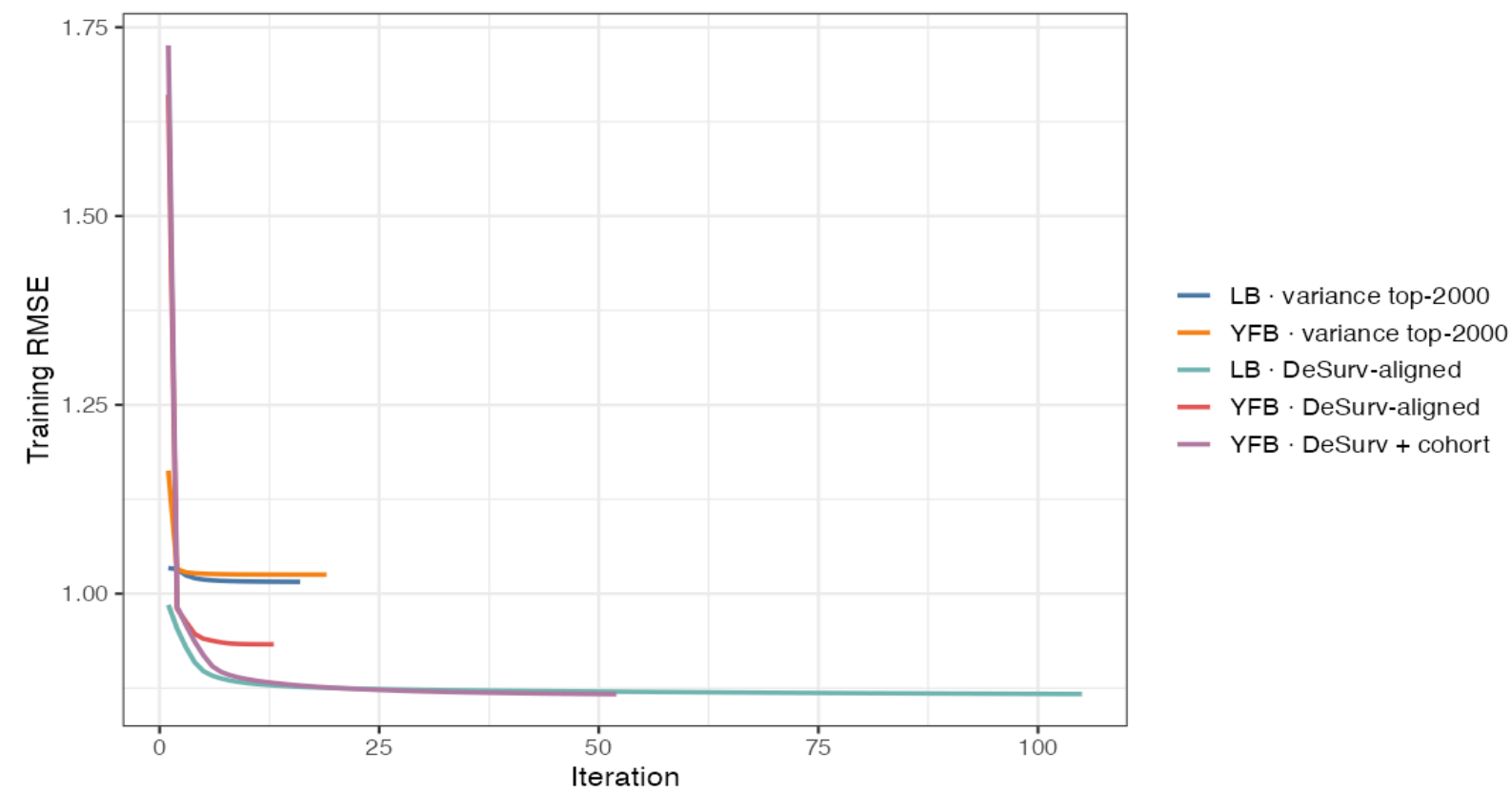


Figure A3. Training RMSE convergence traces across model configurations — RMSE stabilizes to a plateau; stopping is governed by the relative ELBO change falling below 10^{-5} (ΔL , $\Delta \beta$ tracked as diagnostics).

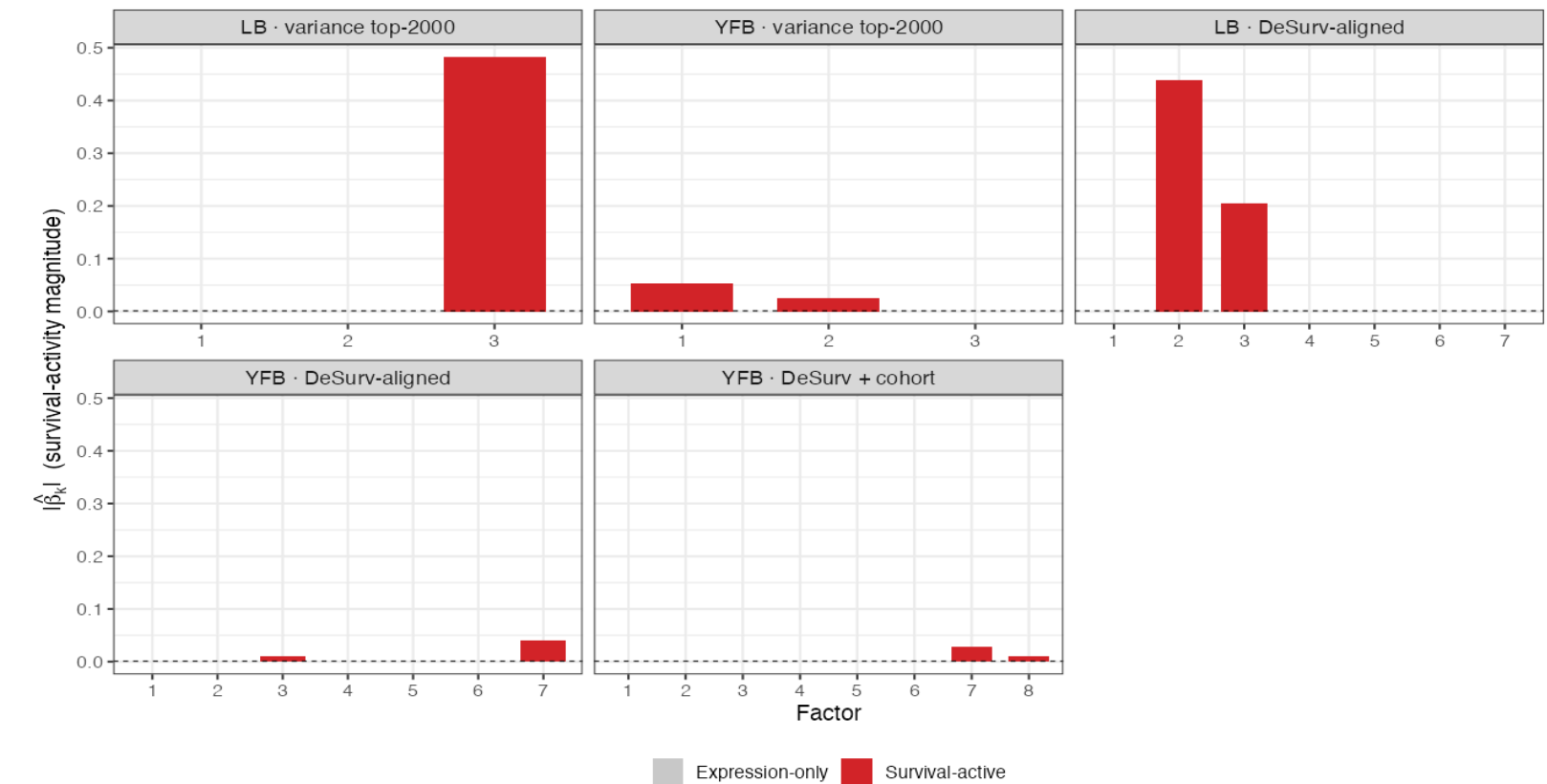


Figure A4. Survival-activity magnitude $|\hat{\beta}_k|$ per program. Most programs fall below the activity threshold (expression-only); a small number clear it (survival-active).

- The spike-and-slab / shrinkage behavior is visible: most programs are shrunk below the survival-activity threshold, with only the survival-active ones retained

Appendix — relation to DeSurv (Young et al., PNAS 2026)

Same design & same data — SBMF is the Bayesian counterpart. Both jointly fit supervised NMF + Cox with **non-negative gene programs**, share the same training set (TCGA + CPTAC, $n = 273$) and the **same five external PDAC cohorts** (Dijk, Moffitt, PACA-AU array/seq, Puleo; $n = 616$), score new patients by **projection** (no retraining), select via **CV C-index + 1-SE**, and use the **combined-rank top-3000-per-cohort** gene selection. Both also find that the **highest-variance program is *not* the most prognostic**.

DeSurv objective: $(1 - \alpha) \mathcal{L}_{\text{NMF}} - \alpha \mathcal{L}_{\text{Cox}}$, with the survival gradient on the gene programs W .

What SBMF changes

	DeSurv	SBMF
Estimation	penalized optimization	Bayesian variational (CAVI)
Tuned by BO	k, α, λ	only K (α, λ not tuned)
Shrinkage	fixed elastic-net	empirical-Bayes priors (learned)
Uncertainty	point estimates	posterior mean + variance
Noise	least-squares	per-gene precision τ ; $E[L^2]$ correction

- **DeSurv:** $k = 3$, $\alpha = 0.334$; pooled external **HR/SD** = 1.50 (95% CI 1.31–1.72)
- **SBMF (YF)β:** $K = 7 \rightarrow K_{\text{eff}} = 2$; mean external **C** = 0.636

Honest caveat: the two report different primary metrics (DeSurv: pooled HR/SD; SBMF: mean external C-index) and use slightly different normalization (DeSurv within-subject rank $\rightarrow$ 1970 genes; SBMF per-platform z-std $\rightarrow$ 2064). A **matched same-protocol head-to-head is a planned next step** — DeSurv is *not* re-run here.